

Zachary Stybel

zachary.stybel.cs@gmail.com | 347-280-8229 | [LinkedIn](#) | [GitHub](#)

EDUCATION

CUNY Baruch College

New York, NY

B.S. in Computer Science — GPA: 3.78

Expected Graduation, May 2027

Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Database Management, Web Design, Web Engineering, Computer Architecture, Discrete Math, Linear Algebra, Calculus 2

TECHNICAL SKILLS

Languages & Frameworks: Python, JavaScript (React.js, Node.js, Express.js), SQL, C++, Java, HTML, CSS, Tailwind CSS, Bootstrap, jQuery, Verilog

Developer Tools: GitHub, Git, CI/CD (Docker, Vercel), Google Agent Developer Kit, Firebase, Postman, Pandas

PROJECTS

Crypto Palace | *React, JavaScript, Docker, Firebase, Vite, Google Charts, Vercel, CoinGecko API*

- Built a **real-time cryptocurrency dashboard** with **React + Vite**, integrating the **CoinGecko API** to display 100+ live market feeds, enabling users to make informed investment decisions.
- Achieved a **100/100 Lighthouse Best Practices score** and enhanced UX with **React hooks** and **Google Charts** for interactive trend analysis, benefitting the **65 active users**.
- Implemented **Firebase Authentication** and **Firestore database** to secure user accounts and sync watchlists across devices
- Containerized the application with **Docker** for consistent local development and streamlined deployment.

StudyPal | *Python, Google ADK, Gemini-2.0 Flash, Streamlit*

- Engineered **agent tools** from scratch (e.g., input parsers, dictionary/list transformers) to convert raw user input into structured data, enabling **Gemini-Flash** agents to run workflows.
- Prototyped a **multi-agent study assistant** coordinating Quiz, Summarizer, and Planner agents to convert lecture notes into quizzes, flashcards, and daily study plans
- Built a **quiz agent** that created 10–20 multiple-choice questions in <5s, **reducing manual prep time by 80%** and improving study efficiency.

Soccer Club Analytics Database | *SQL, MS Access, UML modeling, ER models*

- Led a team in designing a **3NF relational database** schema in MS Access with **7+ interrelated entities**, enforcing referential integrity to ensure data quality across player and club operations.
- Authored **15+ SQL queries and reports** analyzing top spenders, injury trends, and contract expirations, supporting data-driven decision-making for club management.
- Produced detailed **UML diagrams** (ER + class models) to validate schema before implementation, preventing inconsistencies and reducing design errors by **90%**.

Metro Fitness | *JavaScript, HTML, CSS, Bootstrap, jQuery, GitHub*

- Launched a **responsive gym website** on GitHub Pages, optimizing images and scripts to achieve a **99/100 Lighthouse performance** score and enhance cross-browser compatibility.
- Enhanced site responsiveness and layout using Bootstrap 5, CSS Grid, and Flexbox, improving user experience on both desktop and mobile devices.
- Increased **form completion rates by 25%** by implementing jQuery validation with inline error handling to block incomplete submissions.

WORK EXPERIENCE

Exam IQ

New York, New York

Math and English Tutor

Dec 2022 – Jul 2023

- Elevated the academic performance of underperforming students by increasing **failing** test averages to around an **80** average—an improvement of up to **40%**—while maintaining strong outcomes for high achievers.
- Improved student performance through clear explanations, time management strategies, and adaptability to individual learning needs.